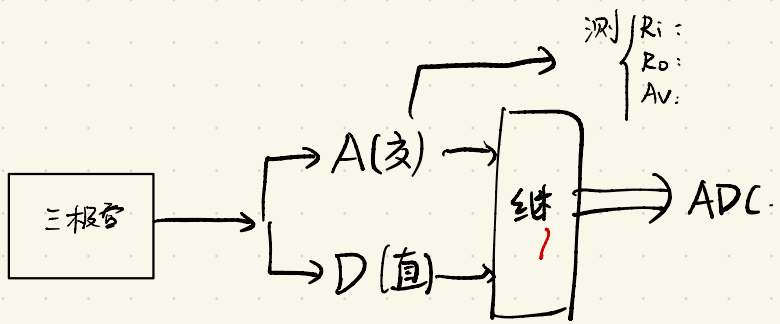
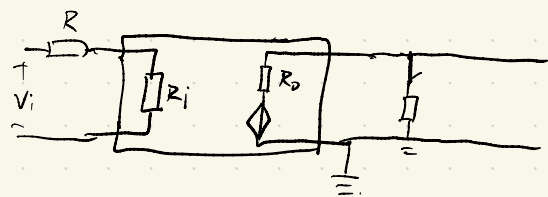


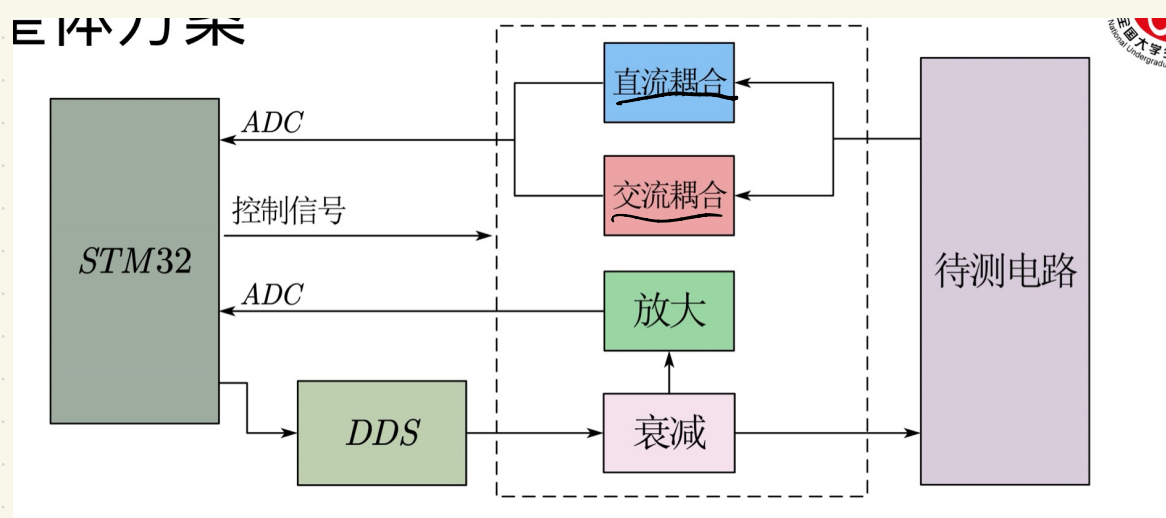




- 交
交
交
交
1. R_i → 一个 GPIO
 2. R_o → 一个 GPIO ⇒ 三极管输出端的 基极 电压
 3. A_v
 4. DDS → A_v
- ⇒ 交流 基极 电压 (A)
计算幅值 ⇒ R_o .
- R接, 测 R 分压
R断, 测 V_o .

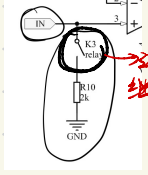


元件列表



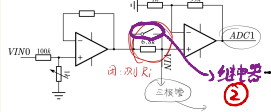
1. 基础

$R_o \Rightarrow \text{DDS out } 1\text{kHz} \Rightarrow$



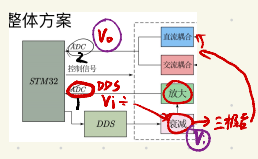
$\left\{ \begin{array}{l} \text{闭合 } R_o \text{ 接入, 控 AD 继电器到 A} \rightarrow \text{ADC 测幅度} \rightarrow V_{oL} \\ \text{断 } R_o \text{ 不接, } \text{---} \text{ 到 A} \rightarrow \text{APL} \text{ ---} \rightarrow V_o \end{array} \right\} \Rightarrow R_o$

$R_i \Rightarrow \text{DDS out } 1\text{kHz}$



$\left\{ \begin{array}{l} \text{闭 } 6.8\text{k} \text{ 短, 控 AD 继电器到 A} \rightarrow \text{ADC 测幅} \rightarrow V_i \\ \text{断 } 6.8\text{k} \text{ 接入, } \text{---} \text{ A} \rightarrow \text{ADC (测)} \rightarrow V_2 \end{array} \right\} \rightarrow R_i$

$A_v = \frac{V_o}{V_i} \Rightarrow \text{DDS out}$



$\left. \begin{array}{l} \text{输入端 R 短 继②闭} \\ \text{+} \\ \text{输出端 R 不接 继④断} \end{array} \right\} \left\{ \begin{array}{l} \text{ADC}_1 \rightarrow \frac{V_i}{\text{放大系数}} \\ \text{ADC}_2 \rightarrow V_o \end{array} \right\} \xrightarrow{\text{20lg}} A_v$

扫频 (不断测 A_v)

2.